AB0732

CELLULAR AND MOLECULAR HALLMARKS OF EARLY METHOTREXATE RESPONSE IN RA

Keywords: Disease-modifying Drugs (DMARDs), '-omics, Biomarkers

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Background: Rheumatoid arthritis (RA) is a chronic inflammatory joint disease that imposes a massive burden on affected individuals and on society. If left untreated, it can lead to irreversible joint damage. While effective treatment options, particularly with methotrexate (MTX) as a first line of treatment, have been used successfully to reduce this burden, multiple challenges still remain [1]. Due to its slow onset of action, treatment with MTX can take up to 6 months before a clinical decision regarding response is made, which can delay the successful treatment in non-responders [1]. In addition, the precise mode of action of MTX in RA on cellular and molecular level still remains unknown.
Objectives: This study aims to elucidate the complex interplay of diverse anti-inflammatory effects on various pathogenic cell types exhibited by MTX over time.
Methods: To explore this, we utilized single-cell RNA sequencing (scRNA-seq) in conjunction with 40 color immunophenotyping [2] of peripheral blood mononuclear cells (PBMCs) obtained from treatment-naive RA patients and the same patients after 3, 6, and 12 weeks of MTX treatment. All patients responded to treatment at the 12 weeks follow up. We performed variance decomposition analysis and gene co-expression network analysis to define cell-specific gene signatures perturbed by MTX. In addition we profiled age- and sex-matched heathy controls, in order to define an RA-specific single-cell landscape at baseline.
Results: Combining the two technologies we detected perturbations of cell proportions in multiple cell types as early as 3 weeks from the start of MTX treatment in RA patients. Variance decomposition analysis showed a large effect of cell heterogeneity on the sample. Stratifying by cell type showed cell-specific effects of the MTX treatment. Shared as well as unique cell-specific early gene signature modules were identified, suggesting an involvement of pro-inflammatory cellular programs in response to MTX in PBMCs.
Conclusion: The combinination of two distinct methods for cellular profiling allows for deep cellular and molecular phenotyping of cells responding to MTX treatment as well as time-resolved insight in the mechanism of action. This, in turn, can be utilized in biomarker development for early MTX response detection before the onset of clinically detectable responses.

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AB0733

EPIGENETIC CONTROL OF GENES AFFECTED BY IMMUNOSUPPRESSIVE TREATMENT IN CD4+ T CELLS OF PATIENTS WITH RHEUMATOID ARTHRITIS

Keywords: Epigenetics, Adaptive immunity, Disease-modifying Drugs (DMARDs)

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Background: Epigenetic processes promote development of pathogenic cell types in rheumatoid arthritis (RA), but precise molecular mechanisms behind these effects remain largely unexplored. Survivin has recently been identified as important regulator of histone deposition on chromatin preventing accumulation of the repressive H3K27me3 mark (1,2).
Objectives: In this study, we investigate if survivin affects acetylation of histone H3K27 in CD4+ T cells, and how this is associated with effect of immunosuppressive RA treatment.
Methods: Chromatin of CD4+ cells (n=12) treated with or without the survivin-inhibitor YM155 was immunoprecipitated with antibodies to histone H3K27ac and sequenced (ChIP-seq, Illumina). Peaks with change >30% in deposition of H3K27ac upon YM155-treatment were annotated to the genomic regulatory elements (RE) via GeneHancer database, which also gave information of genes connected to these RE. Transcriptomics of CD4+ cells isolated from RA patients treated with MTX (n=18), TNFi (n=10), JAKi (n=24) and having no DMARDs (n=7), by RNA sequencing, deposited GSE201669. The genes differentially expressed (DEG, nominal p<0.05) upon JAKi-treatment compared to other treatments were identified by DESeq2 (R-studio, Bioconductor). External transcriptome datasets of CD4+ cells isolated from patients before and after treatment with MTX (n = 28, GSE176440), abatacept (n = 14, GSE121827), and tocilizumab (n = 12, GSE113156). Treatment affected DEG were identified by DESeq2 (R-studio, Bioconductor). The enrichment analysis of DEG for biological processes within Gene ontology library were analyzed through String database. Over-representation analysis was done at http://cpdb.molgen.mpg.de.
Results: Inhibiting survivin with YM155 in CD4+ cells induced a significant (>30%) change in 17% (1967 of 11529 peaks) of H3K27ac-peaks. These survivin-sensitive H3K27ac peaks were located within 339 cis-RE, connected to 1022 protein-coding genes expressed in CD4+ cells. Biological pathway analysis revealed that the top biological processes enriched among the these 1022 genes were mRNA metabolic process (GO:0016071 FDR = 0.0043), Cellular response to DNA damage stimulus (GO:0006974 FDR = 0.0029), DNA metabolic process (GO:0006259 FDR 0.033), RNA processing (GO:0006396 FDR 0.00058), RNA metabolic process (GO:0016070 FDR = 2.70e-05), and Cellular response to stress (GO:0033554 FDR = 0.00099). Together, these processes has a significant overlap and comprised 28% (286 of 1022) of the genes connected to the changed H3K27ac- genes. Immunosuppressive treatment had a significant imprint on H3K27ac connected genes, which were found over-represented among the DEG after treatment with MTX (p = 6e-49, OR = 1.77), abatacept (p = 1.6e-25, OR = 2.31), tocilizumab (p = 1.9e-7, OR = 1.51) and JAKi (p = 0, OR = 3.52) (all by Enrichr: Epigenomics roadmap). Furthermore, we found that JAKi and Mtx affected the same processes of transcription, RNA processing and translation; TGFb and SMAD signaling; NOTCH signaling; and DNA damage response as controlled by the genes connected to changed H3K27ac (JAKi p = 0.010 OR = ∞, Mtx p = 0.0001 OR = 3.2). In the top enriched GO:BP of the genes connected to changed H3K27ac, the DEG of the treatments were enriched to different extent (Table 1). In total, JAKi affected 219, abatacept 17, tocilizubam 17 and MTX 48 of the genes within the top GO:BP.
Conclusion: Immunosuppressive treatment affects genes under the epigenetic control in transcription, RNA processing and DNA damage response exerted through the survivin-sensitive H3K27ac deposition. Molecular signature of CD4+T cells reflecting activation of these processes could assist individual treatment choice in RA patients.

Table 1.

	GO:0016071 p, OR	GO:0006974 p, OR	GO:0006259 p, OR	GO:0006396 p, OR	GO:0016070 p, OR	GO:0033554 p, OR	combined p, OR
JAKi	0.03, 1.4	ns	ns	0.01, 1.4	0.0008, 1.4	ns	0.0008, 1.3
Abatacept	ns	ns	ns	0.007, 3.2	ns	ns	ns
Tocilizumab	ns	ns	ns	ns	ns	0.004, 3.1	0.016, 2.2

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AB0734

PEPTIDYLARGININE DEIMINASE 4 (PAD4) INHIBITOR 4,6-BIS(2,2,2-TRIFLUOROETHOXY)-1,3,5-TRIAZIN-2-AMINE II AMELIORATES COLLAGEN-INDUCED ARTHRITIS IN MICE

Keywords: Animal Models, Cytokines and Chemokines

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Background: Peptidylarginine deiminase-4 (PAD4) is a calcium-dependent enzyme that catalyzes the conversion of arginine into citrulline in macromolecules within the body. Dysregulated PAD4 plays a crucial role in the occurrence and development of Rheumatoid arthritis (RA). Therefore, PAD4 is considered a promising target for diagnosing and treating RA.
Objectives: In this study, we explored for the first time the therapeutic potential of the PAD4 inhibitor known as 4,6-bis(2,2,2-trifluoroethoxy)-1,3,5-triazin-2-amine, referred to as TTA, in collagen-induced arthritis.
Methods: We immunized DBA/1 mice with type II collagen, and the TTA treatment commenced on the 21st day after induction. Throughout the treatment period, we closely monitored alterations in body weight, assessed paw swelling through clinical scoring, and conducted histological examinations of the joints. On the 42nd day, we further examined changes in inflammatory cytokines in the serum and analyzed shifts in T-cell subpopulations within lymphatic tissues using ELISA and flow cytometry techniques.
Results: The findings revealed that TTA treatment could alleviate arthritis symptoms, with improvements observed in joint manifestations and histological damage. Additionally, lung histological staining and experimental analysis of the pro-inflammatory cytokine TNF-α indicated signs of improvement. Furthermore, intracellular staining analysis of the immune system within the inguinal lymph nodes and spleen suggested a reduction in Th17 cells.
Conclusion: In summary, TTA effectively mitigated collagen-induced arthritis in mice, indicating its potential as a future therapeutic option for the treatment of rheumatoid arthritis.

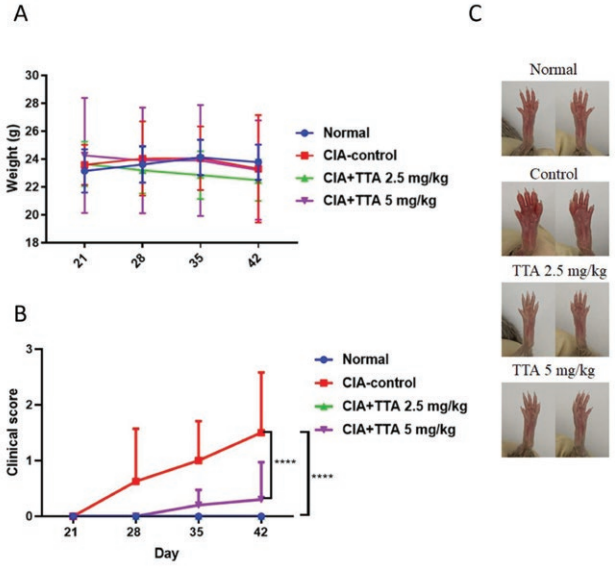


Figure 1. TTA effectively attenuates collagen-induced arthritis symptomsMice with type II collagen-induced arthritis were administered TTA at doses of 2.5 mg/kg and 5 mg/kg daily from day 21 to day 42. A) Weekly body weight records for each group of mice. B) The efficacy of a therapeutic approach can be semiologically quantified using a clinical scoring system. C) Representative images of mouse paws from different groups. (****: p<0.0001)

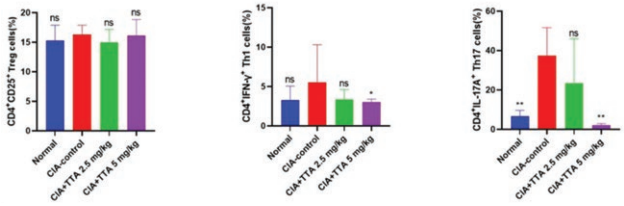


Figure 2. TTA reduces the expression of helper T cells in collagen-induced arthritis mice. Spleen cells from collagen-induced arthritis mice were stained and analyzed by flow cytometry. There was no significant difference in Treg (CD4+CD25+) cells between the treatment group and the control group. However, there was a decreasing trend in Th1 (CD4+IFN-γ+) and Th17 (CD4+IL-17A+) cells after treatment (*: p<0.05, **: p<0.01, ns: not significantly different).

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AB0735 **ALTERED GALECTIN EXPRESSION CORRELATE WITH TNF-MEDIATED INFLAMMATORY CARTILAGE DAMAGE**

Keywords: Animal Models, Biomarkers

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Background: Galectins (Gal) are a family of soluble ß-galactoside-binding proteins that serve as regulators in multiple physiological processes like proliferation or cell adhesion. There is growing evidence that galectins are up-regulated in synovial tissue and sera from patients with RA.1-4 However, the role of galectins in inflammatory bone erosions and cartilage breakdown has not been elucidated in RA, limiting mechanistic insights into their precise function.
Objectives: Thus, the present study investigates the galectin profile in inflammatory joint damage in a TNF-driven arthritis mouse model.
Methods: Articular joints from 10-week old human tumor necrosis factor transgenic mice (hTNFtg, Tg197 strain), with established chronic inflammatory, erosive arthritis model, and from age-matched wild-type (wt) littermates were investigated for the expression of Gal-1, -2, -3, -4, -7, -8 and -9 in tissue extracts and sections. Areas of galectin expression in different compartments of inflamed joints were quantitatively assessed using TissueQuest software. Severity of synovitis, bone erosions and cartilage damage was determined in TRAP and SafO-stained sections and their linkage to individual galectins was calculated by Spearman correlation analysis.
Results: In general, an increased expression of galectin was found in various knee joint compartments such as synovial tissue and menisci of hTNFtg mice compared to wt mice. Progressive joint inflammation led to cartilage damage indicated by complete proteoglycan loss, partial cartilage erosions of the deeper, calcified zone and invasion of the inflammatory synovial pannus tissue. Interestingly, immunohistochemical stainings revealed an increased expression of Gal-1, -4, -7 and -8, but not of Gal-2, -3 and -9 in cartilage from hTNFtg mice compared to healthy cartilage from wt animals. These findings were observed particularly in tibial rather than femoral cartilage. Whereas most galectins were found to be exclusively expressed in chondrocytes, Gal-1, -4 and -7 overexpression was also apparent in cartilage matrix of hTNFtg mice. Furthermore, correlation analysis revealed a significant linkage between increased galectin-1, -4, -7, -8 expression and progression of cartilage erosion. In contrast, Gal-9 was found to be downregulated upon damage.
Conclusion: Under TNF-driven inflammatory conditions, a distinct pattern of glycan-binding galectins is notable in articular cartilage. The change in galectin pattern is linked to the progression of cartilage erosion, which suggests a functional role of galectins in the pathophysiology of inflammatory cartilage damage.
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AB0736 **DEVELOPMENT AND APPLICATION OF EPITOPESCAN, A PYTHON3 TOOLSET FOR MUTATION TRACKING IN SARS-CoV-2 IMMUNOGENIC EPITOPES RESTRICTED BY THE RHEUMATOID ARTHRITIS SUSCEPTIBILITY ALLELE HLA-DRB1*04:01**

Keywords: '-omics, Adaptive immunity, Genetics, Artificial Intelligence, Vaccination/Immunization

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